

ELEC 244 Spring 2024

Final Lab Report

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1 Objective

The purpose of this final lab experiment was for the students to culminate their knowledge of audio amplifiers and filtering to build an analog spectrum audio analyzer.

2 Test Description

2.1 Pre-Lab Calculations

To calculate the low and high cutoff frequencies for each of the bandpass filters, the students will use the following equation.

$$f_c = \frac{1}{2\pi RC}$$

For the top bandpass filter, which is our high-frequency bandpass, the equation for our low and high cutoffs becomes

$$f_{c, low} = \frac{1}{2\pi(0.1\mu F)(100\Omega)} \approx 16\text{kHz},$$
$$f_{c, high} = \frac{1}{2\pi(330\Omega)(0.1\mu F)} \approx 4.8\text{kHz}$$

For the middle bandpass filter or our mid-frequency bandpass, the equations become

$$f_{c, low} = \frac{1}{2\pi(0.015\mu F)(4.7k\Omega)} \approx 2.25\text{kHz},$$
$$f_{c, high} = \frac{1}{2\pi(630\Omega)(1\mu F)} \approx 253\text{Hz}$$

For the bottom bandpass filter or our low-frequency bandpass, the equations become

$$f_{c, low} = \frac{1}{2\pi(0.015\mu F)(100k\Omega)} \approx 106\text{Hz},$$
$$f_{c, high} = \frac{1}{2\pi(10k\Omega)(0.015\mu F)} \approx 16\text{Hz}$$

2.2 Audio Amplifier

First, students will construct an audio amplifier circuit as pictured in Fig. 1, using a LM386 op amp. Students should use values of $C = 1\mu\text{F}$ and $4.7k\Omega$, 470Ω , $100\mu\text{F}$, $10\mu\text{F}$. Once they have constructed the circuit, students should note the output of a pure tone signal in the oscilloscope.

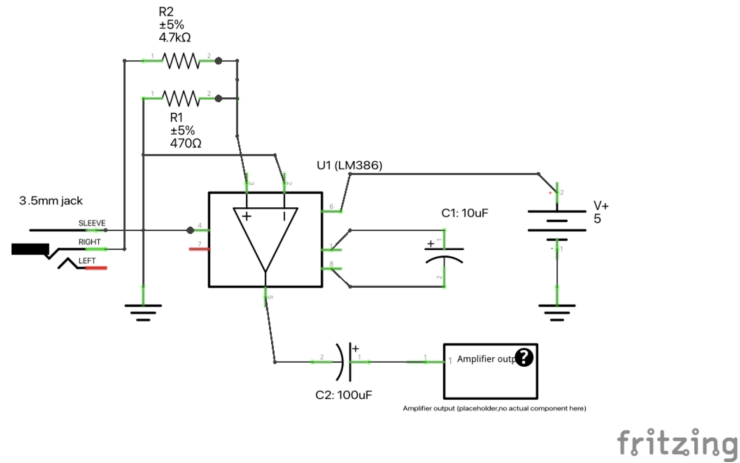


Figure 1: Audio amplifier schematic.

The students will then play music through the aux connected to the audio amplifier circuit. To verify your audio amplifier circuit is correct, make sure that when you pause your music, the waveform disappears.

2.3 Filtering

Next, students will construct three bandpass filters which each operate on the output of the audio amplifier as pictured in Fig. 2. The students will use three of the $22\text{ k}\Omega$ resistors, three of the $390\text{ k}\Omega$ resistors, three of the OP07 op amps, two of the $0.1\text{ }\mu\text{F}$ Capacitor, one $100\text{ }\Omega$ resistor, $630\text{ }\Omega$ resistor, and four of the $330\text{ }\Omega$ resistors, one $10\text{ k}\Omega$ resistor, two of the $0.015\text{ }\mu\text{F}$ capacitors, one $100\text{ k}\Omega$ resistor, one $4.7\text{ k}\Omega$. The top bandpass filter will be the high-frequency band, the middle will be the middle-frequency band, and the low will be the low-frequency band.

To test that the bandpass filters work as expected, the students will connect to each output of the OP07 op amps, a $330\text{ }\Omega$ resistor in series with an LED. The students will then sweep through a band of pure tones and note the behavior of the three LEDs.

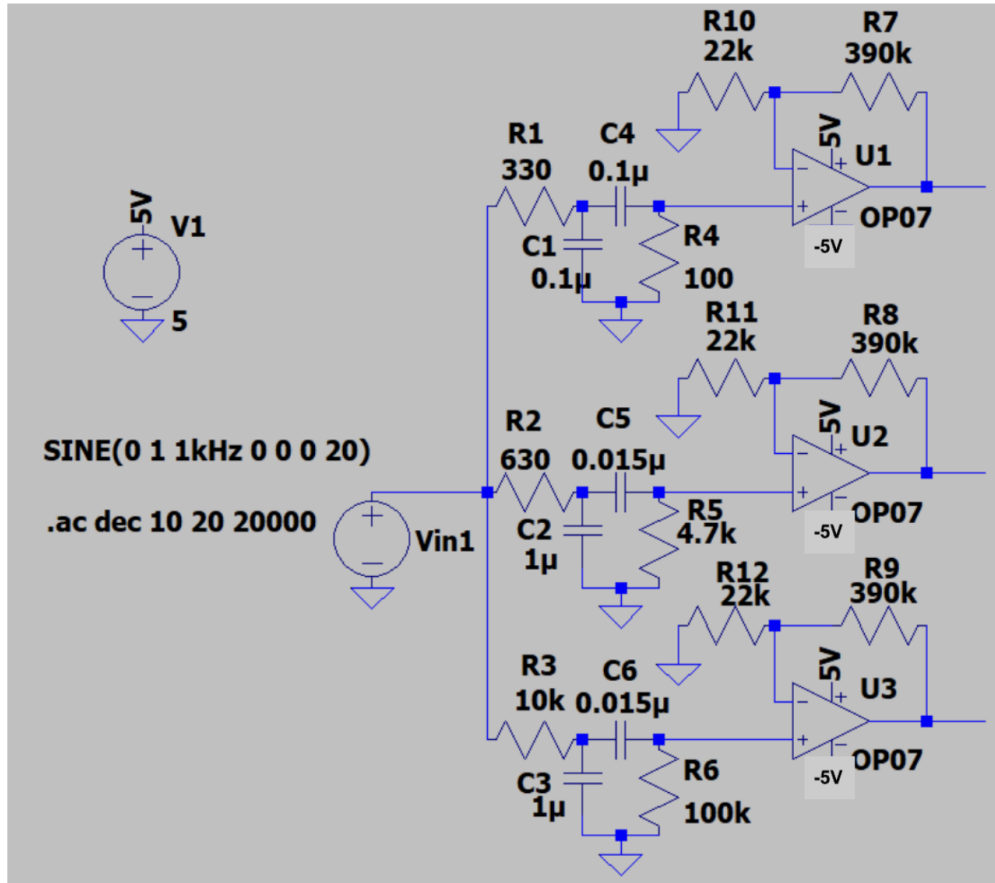


Figure 2: Schematic of three bandpass filters.

2.4 Output

The students will now connect the output of each of the bandpass filters to an instantiation of the circuit shown in Fig. 3.

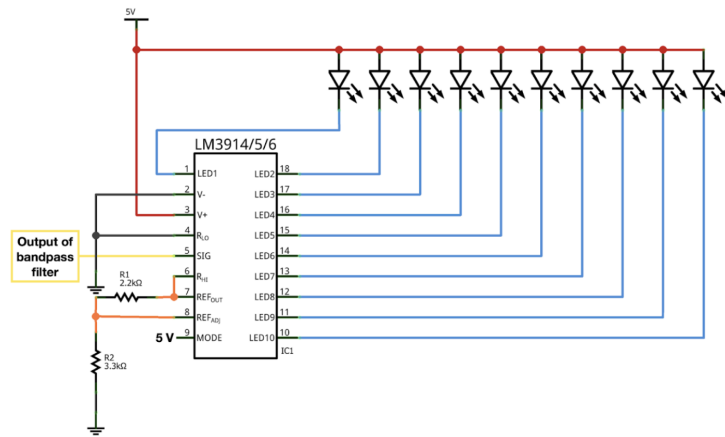


Figure 3: LM3914 and LED strip output schematic.

2.5 Bill of Materials

- National Instruments VirtualBench and associated cables
- PC and associated cables
 - Software required: VirtualBench
- Breadboard
- Banana cables (1x red, 1x green)
- 3x BNC cables
- Circuit components:
 - 2x 4.7 k Ω Resistor
 - 3x 2.2 k Ω Resistor
 - 3x 3.3 k Ω Resistor
 - 1x 470 Ω Resistor
 - 1x 630 Ω Resistor
 - 1x 100 Ω Resistor
 - 1x 100 k Ω Resistor
 - 1x 10 k Ω Resistor
 - 4x 330 Ω Resistor
 - 3x 22 k Ω Resistor
 - 3x 390 k Ω Resistor
 - 1x 10 μ F Capacitor
 - 2x 1 μ F Capacitor
 - 2x 0.1 μ F Capacitor
 - 2x 0.015 μ F Capacitor
 - 1x 100 μ F Capacitor
 - 3x OP07 Op Amp
 - 3x LM3914 Op Amp
 - 1x LM386 Op Amp
 - 1x Red LED
 - 1x Green LED
 - 1x Yellow LED
 - 1x 3.5mm Aux Cable
 - 3x LED Strips
- Additional colored wires, as needed for breadboard and circuit setup

2.6 Procedure

1. Set up the breadboard.
 - (a) Connect horizontal breadboard rails to power terminals (paired to binding posts).
 - (b) (Optional) Connect PN diodes in reverse across the power rails ($+$ $\rightarrow$ GND, GND $\rightarrow -$).
 - (c) Using 3 banana cables of the appropriate colors, connect the binding posts of the breadboard to the power supply of the VirtualBench mixed signal oscilloscope. Connect the green cable from the VB GND (hollow triangle) to the breadboard ground, the red cable from the $+15\text{V}$ breadboard binding post to the $+25\text{V}$ on the VB, and the black cable from the -15V breadboard to the -25V on the VB.
2. Build the audio amplifier.
 - (a) Use the LM386 Op Amp, the 3.5mm aux cable input, the $10\ \mu\text{F}$ Capacitor, and the $100\ \mu\text{F}$ Capacitor to construct the audio amplifier shown in Fig. 1. The power supply voltage should be $+5\text{V}$.
 - (b) Use the oscilloscope to see the amplifier output with music playing through the 3.5mm cable connected to the breadboard.
3. Build the three bandpass filters.
 - (a) Using three of the $22\ \text{k}\Omega$ resistors, three of the $390\ \text{k}\Omega$ resistors, three of the OP07 op amps, two of the $0.1\ \mu\text{F}$ Capacitor, one $100\ \Omega$ resistor, $630\ \Omega$ resistor, and four of the $330\ \Omega$ resistors, one $10\ \text{k}\Omega$ resistor, two of the $0.015\ \mu\text{F}$ capacitors, one $100\ \text{k}\Omega$ resistor, one $4.7\ \text{k}\Omega$ resistor, to construct the three bandpass filters that will be added in series to the output of the audio amplifier. Use ?? as a reference. Use the -5V and $+5\text{V}$ power rail to power the op-amps.
 - (b) Using the oscilloscope, show the capacitor charging/discharging waveform on CH1, and the output of the CMOS configuration on CH2.
4. Construct the output circuit.
 - (a) Using the rest of the materials, construct the output circuit shown in Fig. 2.
 - (b) Repeat this three times for each frequency band.
 - (c) Play some music!

***Procedure derived from Dr.Young's Final Lab:** Final Lab Procedure

3 Results and Discussion

3.1 Stage 1: Amplification

Figure 4 shows the students' completed amplifier circuit, and Fig. 5 shows the output observed from this circuit using the Virtual Bench oscilloscope when music was provided as an input.

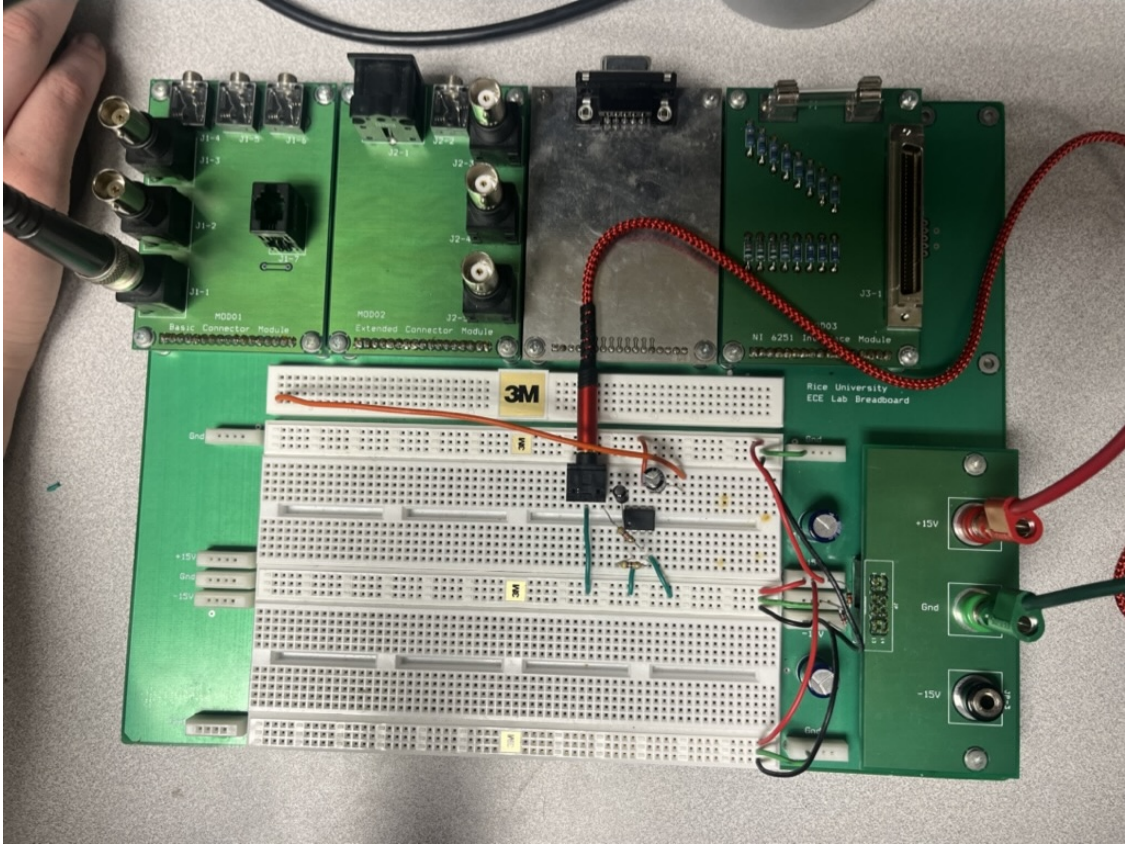


Figure 4: Completed amplifier circuit.

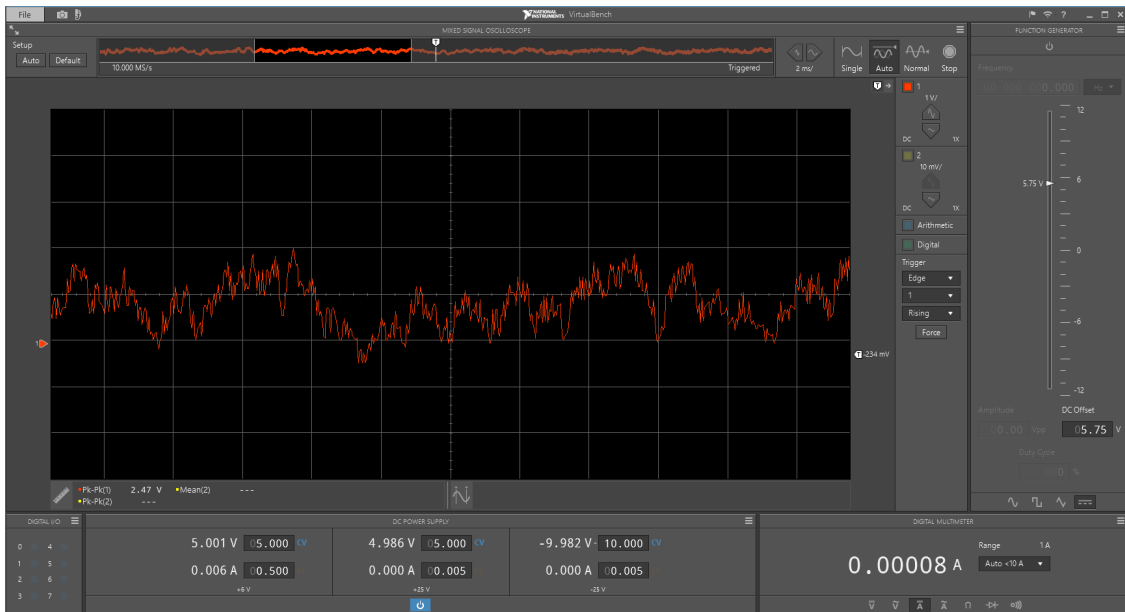


Figure 5: Amplifier output with music as input.

3.2 Stage 2: Filtering

Figure 6 shows the students' completed filter circuit. When provided with pure tone inputs, the LEDs lit up based on the frequency band in the desired manner.

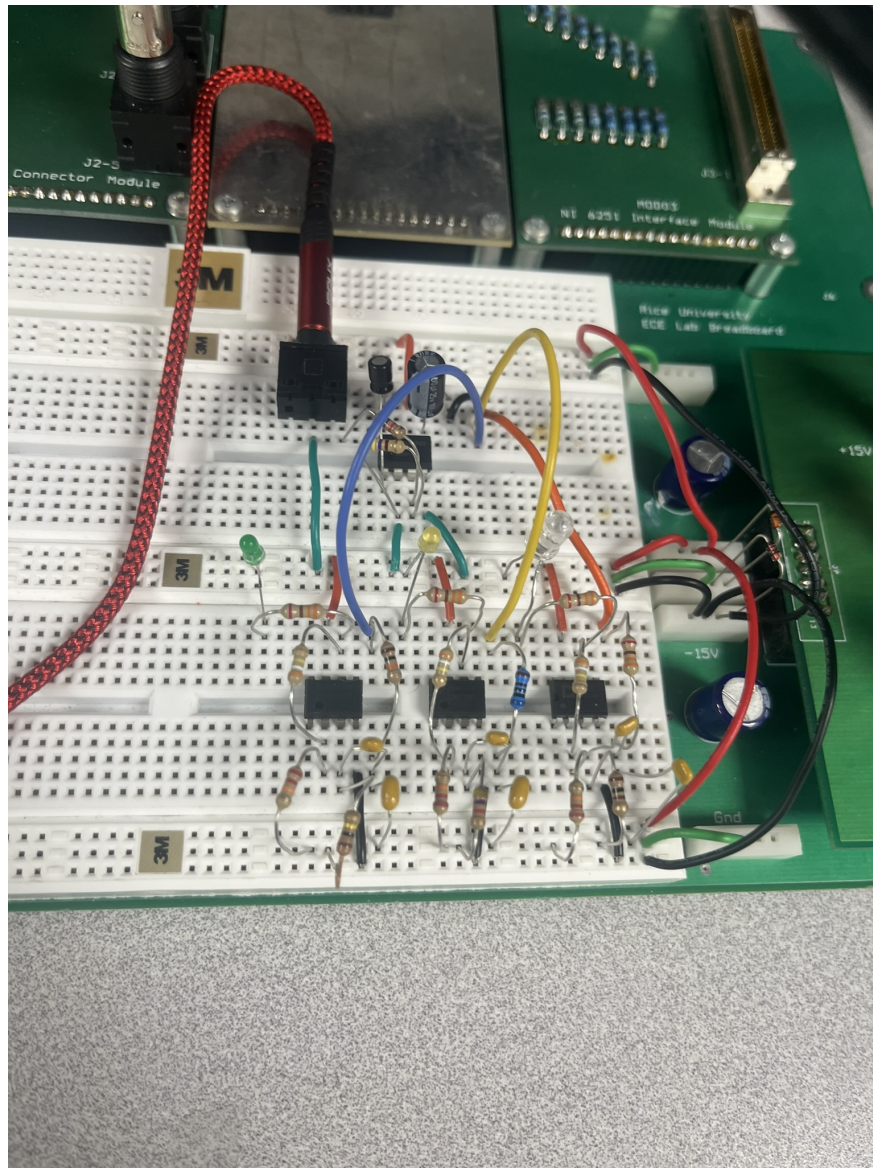


Figure 6: Completed filter circuit.

3.3 Stage 3: Output

Figure 7 shows the students' completed filter circuit. When provided with music input inputs, the LED bars lit up to indicate the frequencies present in the music.

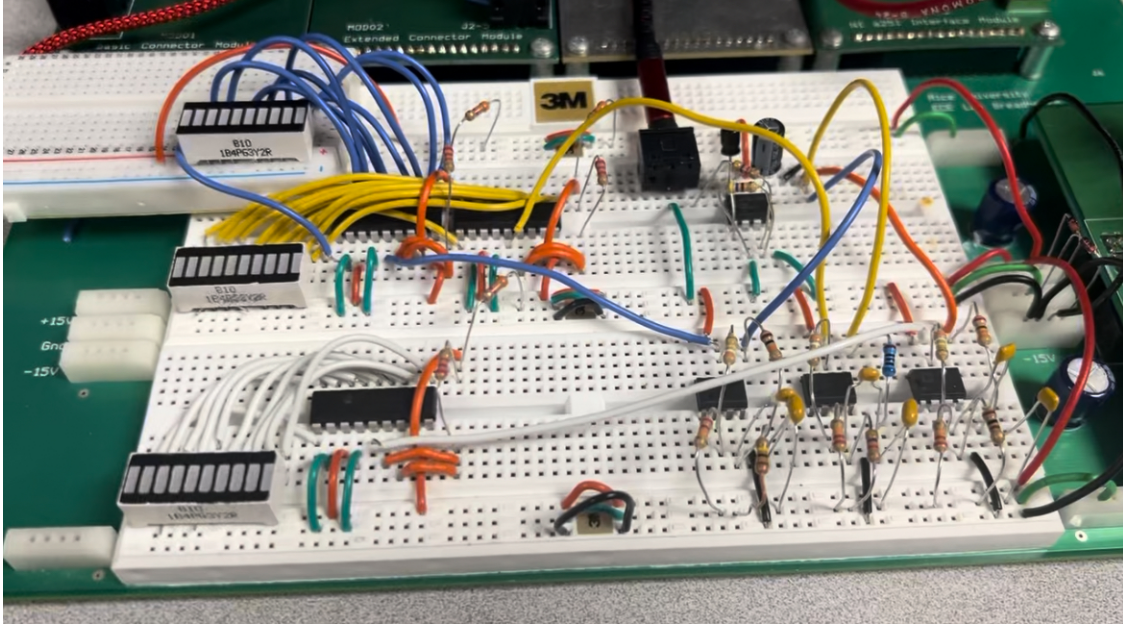


Figure 7: Completed output circuit.

3.4 Discussion

The students observed the desired behavior of the constructed circuits in all parts of the lab. For the amplifier, the circuit was able to amplify the provided input, making the waveform visible on the oscilloscope. For the filtering circuit, the filters were able to output voltage based on the frequencies present in the provided audio. Finally, the output circuit worked as desired, turning on LEDs to indicate the frequencies in the music.

4 Conclusion

Reiterating what was previously written in section 1, the students were able to successfully produce an audio analyzer. This helped them develop skills that are used in many areas of electronic development.